

Juplaya Trailer Power Lead Calculations

Planning worksheet for the TikZ power overview. This is not a final wiring schematic or code sign-off. Fuse class, interrupt rating, insulation type, conduit/bundle derating, crimp hardware, strain relief, and exact route lengths still need the measured install drawing.

Inputs and Hot-Day Assumptions

Item	Value used
Primary sources	Project source files listed below the assumptions table; numeric values come from the current power design and local component spec notes.
Peak environment	Hot summer design ambient: 46 C / 115 F. Voltage-drop calculations use 75 C conductor resistance, so the assumed conductor rise for the drop calculation is $\Delta T = 29$ C above ambient.
Copper temperature correction	$R_T = R_{20}[1 + 0.00393(T - 20)]$, with $T = 75$ C.
Voltage drop	$V_{drop} = 2LIR_T/1000$, where L is one-way length in ft and R_T is ohms per 1000 ft at 75 C.
PV module current	LG455N2W-E6: $I_{mpp} = 10.83$ A and $I_{sc} = 11.39$ A. Series strings keep the same current.
Charge-current convention	PV-string labels show string current. MPPT-to-battery labels show approximate 48 V charge current and controller output limit.
Wire-size convention	“Minimum / plan” means the smallest conductor selected for this planning model after ampacity and voltage-drop screening. It is not permission to ignore OCP, terminal temperature, bundling, conduit fill, or manufacturer instructions.

Source files: `docs/power.md`; LG455, Victron SmartSolar, Anker C1000/PS400, LiTime ComFlex, Dometic CFX3, and Orion reference notes under `docs/reference/`; Velit current range from `docs/research/aio-inverter-3panel-evidence-2026-06-05.md`.

Peak Current Notes

- Roof PV: 3×455 W = 1365 W. Approximate 48 V charge current is 1365 W / 48 V = 28.4 A; at 57.6 V absorption it is 23.7 A. The SmartSolar 250/60-Tr output limit is 60 A.
- Optional LG ground PV: 2×455 W = 910 W. Approximate 48 V charge current is 19.0 A at 48 V and 15.8 A at 57.6 V. The SmartSolar 150/35 output limit is 35 A.
- Velit 2000R: source notes carry about 5–14 A at 48 V, rated about 670 W. Use 14 A for branch voltage-drop sizing until the final manual/fuse value is confirmed.
- 24 V bus: the Orion-Tr 48/24-16A is the hard rail limit. Fridge + normal lighting + USB fits; all floods, full awning, USB-C full load, optional C1000 top-up, and heater glow do not fit simultaneously.
- Phase 2 MultiPlus is deferred. If AC charging is added later, combined trailer charging must still be capped at or below the 100 A ComFlex charge limit.

Lead Sizing Worksheet

Lead	Current basis	Circuit V	Assumed one-way length	Required ampacity basis	Minimum / plan conductor	Worst-case drop at 75 C	Notes
Roof 3S PV string to SmartSolar 250/60-Tr	10.83 A MPP; 11.39 A Isc	126.3 Vmp	20 ft	PV conductor and OCP at or above 20 A module series-fuse class	10 AWG / 2.59 mm	0.53 V / 0.4%	3S raises voltage; current stays one-panel current.
Optional LG ground 2S string to SmartSolar 150/35	10.83 A MPP; 11.39 A Isc	84.2 Vmp	50 ft	PV conductor and OCP at or above 20 A module series-fuse class	10 AWG / 2.59 mm	1.32 V / 1.6%	Portable length assumption; measure actual deployable lead.
SmartSolar 250/60-Tr to 48 V bus	24–28 A expected; 60 A controller limit	57.6 V charge	4 ft	60 A controller output	6 AWG / 4.12 mm	0.23 V / 0.4%	Size OCP to controller and conductor, not average harvest.
SmartSolar 150/35 to 48 V bus	16–19 A expected; 35 A controller limit	57.6 V charge	4 ft	35 A controller output	8 AWG / 3.26 mm	0.21 V / 0.4%	Optional source, but full-output conductor basis.
Battery main OCP / shunt / 48 V bus	100 A continuous pack limit	51.2 V nominal	3 ft	100 A charge/discharge limit, plus interrupt-rating gate	2 AWG / 6.54 mm	0.11 V / 0.2%	Main OCP class and interrupt rating remain open.
48 V bus to Velit 2000R branch	5–14 A operating range	48 V nominal	15 ft	At least 14 A plus final Velit OCP value	12 AWG / 2.05 mm	0.81 V / 1.7%	Use final Velit manual/fuse value when available.
48 V bus to Orion 48/24 input	About 9.2 A at 384 W output and 87% assumed efficiency	48 V nominal	4 ft	20 A input breaker class from build notes	12 AWG / 2.05 mm	0.14 V / 0.3%	Efficiency is assumed pending exact 48/24 datasheet row.
Orion 24 V output to Blue Sea 5026	16 A converter output limit	24 V regulated	3 ft	16 A rail output	12 AWG / 2.05 mm	0.19 V / 0.8%	Converter is the bus bottleneck.
Fridge branch	4.6 A running	24 V regulated	18 ft	10 A branch fuse; 3% voltage-drop target	14 AWG / 1.63 mm	0.51 V / 2.1%	Matches current build note if one-way run stays near 18 ft.
Longest lighting/flood trunk	8 A planning trunk load	24 V regulated	35 ft	5 A per branch; trunk must match aggregate branch plan	10 AWG / 2.59 mm	0.68 V / 2.8%	Long exterior runs drive the size; local zones may be smaller.
USB-C PD branch	5 A full USB-C load	24 V regulated	12 ft	7.5 A branch fuse	14 AWG / 1.63 mm	0.37 V / 1.5%	Full USB-C load is a short-duration worst case.
Optional C1000 top-up from 24 V bus	10 A C1000 low-voltage input limit	24 V regulated	8 ft	10 A manual/fused branch	12 AWG / 2.05 mm	0.31 V / 1.3%	Discretionary; shed before it starves fridge/house loads.
Winter heater rough-in	1–2 A run; up to 11 A glow estimate	24 V regulated	25 ft	15 A rough-in branch	10 AWG / 2.59 mm	0.67 V / 2.8%	Docs mention 12 AWG; 10 AWG is the voltage-drop planning pick at this assumed length.
48 V bus to Orion 48/12 input	5.8 A at 240 W output and 87% efficiency	48 V nominal	4 ft	Input OCP sized to converter and conductor	14 AWG / 1.63 mm	0.14 V / 0.3%	12 V receptacles stay cabinet-local only.
12 V cabinet receptacle bank	20 A / 240 W converter output	12.2 V regulated	3 ft	20 A converter output, branches fused individually	12 AWG / 2.05 mm	0.23 V / 1.9%	Keep physically short; do not distribute a 12 V house rail.
Anker PS400 to C1000	8.33 A operating; C1000 high-range input allows 12.5 A	48 V operating	15 ft	Anker cable/input rating	10 AWG / 2.59 mm	0.30 V / 0.6%	Separate portable island; never combine with LG strings.

Load-Shedding Implications

The 24 V rail cannot be sized by summing every optional load and pretending they are simultaneous. A literal worst case is fridge 4.6 A + interior lights 3 A + all exterior floods 5.25 A + awning strip 2.7 A + USB-C 5 A + C1000 top-up 10 A = 30.55 A before heater glow. That is almost twice the 16 A Orion limit. The operating rule is therefore part of the electrical design: C1000 top-up, all-floods mode, full USB-C load, and heater glow are mutually managed loads.

The roof and ground MPPT outputs are the opposite case: normal current is below controller capacity, but the conductor/OCP basis should follow controller output limits. Roof plus optional ground charging at expected full-array output is roughly 40–47 A depending on battery voltage. If both controllers are allowed to their hardware limits, they can contribute up to 95 A, still below the ComFlex 100 A charge cap, but leaving no room for a future AC charger.

Open Measurements Before Final Wiring

- Replace every assumed one-way length with measured route length after the nose cabinet, roof gland, Velit station, fridge bay, switch panel, and exterior lighting paths are fixed.
- Confirm conductor insulation temperature rating, strand class, terminal temperature limits, conduit or loom bundling, and whether any wire runs share hot cabinet air.
- Choose final 48 V OCP hardware with adequate DC interrupt rating. The main battery OCP remains a Class-T-or-equivalent gate.
- Bench-measure SmartScan and winter-heater glow current before making them part of a simultaneous-load case.
- After shakedown, measure voltage at the load while hot and loaded; the calculated drops are only as good as the assumed lengths and terminations.